

Discrete Settlement Tensors and Constrained Optimization in DATA 3.0 / AMLD

Settlement Density, One-Hot Terminal States, and Resource Allocation

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Abstract

A multi-agent conjecture settler must decide where finite computation should go before any terminal certificate has been found. This note distinguishes actual settlement from predictive navigation quantities, introduces a discrete settlement tensor for direct, lemma, and cross-agent value, and formulates resource allocation as a constrained optimization problem. Under standard compactness and concavity assumptions, optimal allocations exist and KKT conditions characterize global optima. Strict or strong concavity gives uniqueness and stability. A simple diminishing-returns model yields an explicit water-filling scheduler. Uniform error in an estimated settlement objective produces a quantitative 2ϵ near-optimality guarantee. These results address a meta-level optimization problem; they do not imply that arbitrary mathematical targets are decidable.

AI Integrity Statement

Brian Tenneson is the author of this manuscript and is responsible for its mathematical claims, interpretation, and release. OpenAI ChatGPT was used as an AI research and drafting consultant for mathematical formalization, literature search, exposition, figure design, and LaTeX preparation. No theorem or experimental claim is accepted solely because it was suggested by an AI system. Formal claims are intended to stand on explicit definitions, assumptions, and proofs. Experimental claims require frozen protocols, retained search data, and verifier-backed certificates. Proposed AMLD terminology and architecture remain research proposals unless independently validated.

1 Actual settlement is mutually exclusive

For a target w over a frozen base theory T , let

$$\mathcal{C}_P, \quad \mathcal{C}_R, \quad \mathcal{C}_I, \quad \mathcal{C}_C$$

be terminal certificate classes for proof, refutation, independence, and contradiction. The contradiction class means that both w and $\neg w$ have been certified in the declared base/context. The terminal classes are required to be pairwise disjoint.

Define certified settlement indicators

$$S_k(x) = \mathbf{1}\{x \in \mathcal{C}_k\}, \quad k \in \{P, R, I, C\}.$$

Then

$$\boxed{S_P(x) + S_R(x) + S_I(x) + S_C(x) \leq 1.}$$

At a settled state exactly one coordinate is 1; before settlement all four actual-settlement indicators are 0. Fractional quantities such as F_P, F_R, F_I, F_C are predictions or navigation measurements, not terminal states.

2 What settlement density means

The phrase *settlement density* is proposed terminology. It means the weighted amount of certificate-relevant verified information found per unit search volume or computational cost. A region containing many easy but irrelevant lemmas can have lower settlement density than a sparse region containing one lemma that sharply reduces the remaining proof, refutation, or independence horizon.

A schematic local density for certifier k is

$$\rho_k(x) = \frac{\sum_{y \in N(x)} w_k(y) \mathbf{1}\{y \text{ is verified and useful to } k\}}{\text{Vol}(N(x))}.$$

The weight $w_k(y)$ measures settlement relevance rather than mere theoremhood.

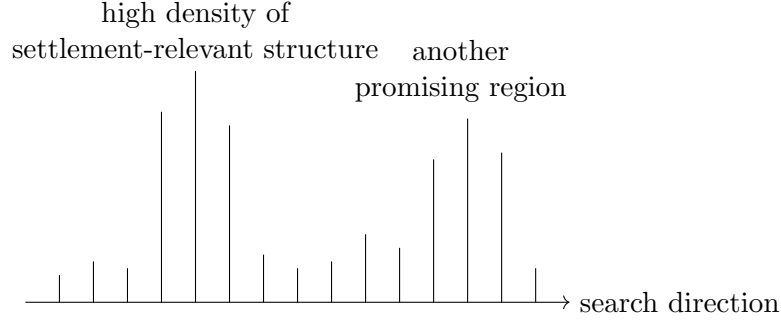


Figure 1: Settlement density is not ordinary statement density. It weights verified discoveries by relevance to eventual certification.

3 The discrete settlement tensor

A scalar density loses multi-agent structure. Let

$$\mathcal{S}_{a,b,r,c}(x)$$

be a state-dependent tensor indexed by source component a , recipient certifier or agent b , search region/frontier class r , and terminal certificate class $c \in \{P, R, I, C\}$. A useful decomposition is

$$\mathcal{S} = \mathcal{S}^{\text{direct}} + \alpha \mathcal{S}^{\text{lemma}} + \beta \mathcal{S}^{\text{transfer}}.$$

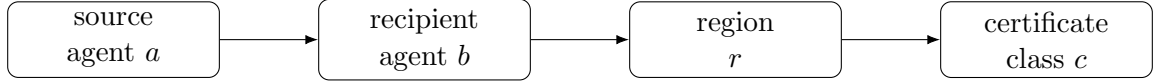
The transfer term permits a discovery made by P , for example, to have low immediate proof value but high value for I through a shared verified bank.

4 Constrained resource allocation

Let $u = (u_1, \dots, u_n)$ be computational allocations and let B be total budget. Let $F_x(u)$ be a calibrated settlement objective inferred from the current state and tensor. The simplest problem is

$$\max_{u \in \mathcal{U}} F_x(u), \quad \mathcal{U} = \{u : u_i \geq m_i, \sum_i u_i = B, Au \leq b\}.$$

The lower bounds m_i can implement anti-starvation requirements for P, R, I , or other essential components.



$$\mathcal{S}_{a,b,r,c}(x)$$

Figure 2: One tensor entry asks who generated a discovery, who can use it, where it occurred, and which mutually exclusive terminal class it may advance.

Proposition 4.1 (Existence). *If $\mathcal{U}$ is nonempty and compact and F_x is continuous, then an optimal allocation exists.*

Proof. This is the extreme-value theorem applied to F_x on the compact feasible set $\mathcal{U}$. $\square$

Ignoring inequalities momentarily, the Lagrangian for $\sum_i u_i = B$ is

$$\mathcal{L}(u, \lambda) = F_x(u) - \lambda \left(\sum_i u_i - B \right).$$

At an interior optimum,

$$\frac{\partial F_x}{\partial u_i} = \lambda.$$

Thus every actively funded channel has equal marginal expected settlement value at the optimum. The multiplier λ is the shadow value of an additional unit of total computation. With inequality constraints the appropriate first-order framework is KKT [1, 2].

Theorem 4.2 (Global KKT optimality under concavity). *Assume $\mathcal{U}$ is convex, F_x is differentiable and concave, and a suitable constraint qualification such as Slater's condition holds. Then every feasible KKT point is a global maximizer of F_x . If F_x is strictly concave, the maximizer is unique.*

5 An explicit diminishing-returns scheduler

Consider

$$F(u) = \sum_{i=1}^n w_i (1 - e^{-a_i u_i}), \quad w_i, a_i > 0.$$

Each summand is increasing and strictly concave. For an interior active channel,

$$w_i a_i e^{-a_i u_i^*} = \lambda,$$

so

$$u_i^* = \frac{1}{a_i} \log \left(\frac{w_i a_i}{\lambda} \right).$$

With lower bounds one clips this expression at m_i , and chooses λ so that the budget constraint holds. This is a concrete water-filling-style scheduler.

6 Quadratic tensor interactions and curvature

A useful local model is

$$F_x(u) = s(x)^T u - \frac{1}{2} u^T Q(x) u.$$

If $Q(x) \succeq 0$, then $\nabla^2 F_x = -Q(x) \preceq 0$ and the allocation problem is concave. If $Q(x) \succ 0$, the optimum is unique. Strong positive cross-agent complementarities can make the Hessian indefinite, in which case KKT no longer guarantees a global optimum. The spectrum of the settlement Hessian is therefore a diagnostic for whether a current allocation problem lies in an easy concave regime or a harder nonconvex regime.

7 Robustness to measurement error

The true settlement objective is not observed directly. Suppose $\hat{F}$ is an estimated objective satisfying

$$\sup_{u \in \mathcal{U}} |\hat{F}(u) - F(u)| \leq \varepsilon.$$

Let u^* maximize F and let $\hat{u}$ maximize $\hat{F}$.

Theorem 7.1 (2ε near-optimality).

$$\boxed{F(u^*) - F(\hat{u}) \leq 2\varepsilon.}$$

Proof. Because $\hat{u}$ maximizes $\hat{F}$,

$$\hat{F}(\hat{u}) \geq \hat{F}(u^*).$$

Uniform error then gives

$$F(\hat{u}) \geq \hat{F}(\hat{u}) - \varepsilon \geq \hat{F}(u^*) - \varepsilon \geq F(u^*) - 2\varepsilon.$$

□

If F is μ -strongly concave, then

$$F(u^*) - F(\hat{u}) \geq \frac{\mu}{2} \|\hat{u} - u^*\|^2,$$

so

$$\boxed{\|\hat{u} - u^*\| \leq 2\sqrt{\varepsilon/\mu}.}$$

Thus a sufficiently well-calibrated imperfect tensor can still yield a quantitatively near-optimal resource allocation.

8 Dynamic allocation

Every verified discovery can change the bank, frontiers, and cross-agent value. Therefore F_t and the optimal allocation u_t^* move with time. A simple projected online update is

$$u_{t+1} = \Pi_{\mathcal{U}}(u_t + \eta \nabla F_t(u_t)).$$

Under bounded-domain and bounded-gradient assumptions, online gradient methods obtain sublinear static regret [3]. If the optimum itself moves slowly, dynamic-regret analyses compare the controller to a moving benchmark.

For computation allocated one discrete quantum at a time, adaptive greedy policies become relevant when the expected utility satisfies adaptive monotonicity and adaptive submodularity [4].

9 Connection to the proof-ocean compass

The tensor is not itself the final navigator. One useful hierarchy is

$$\text{local density and landmarks} \longrightarrow \mathcal{S}(x) \longrightarrow \hat{d}_{\mathcal{C}}(x) \longrightarrow \text{compass} \longrightarrow \text{next tunnel}.$$

On solved proof DAGs, exact distance to a terminal certificate provides an objective training label. Learning-guided ATP and progress-prediction work provide evidence that prior proofs and estimates of remaining proof progress can guide search [5, 6].

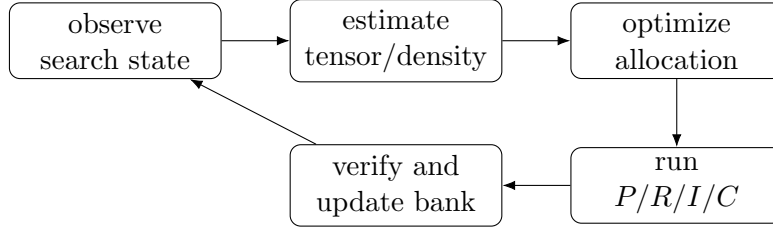


Figure 3: Closed-loop control. The optimizer chooses where computation goes; the verifier alone determines whether a mutually exclusive terminal class has actually been reached.

10 What these results do and do not establish

The results above conditionally solve parts of the AMLD resource-allocation meta-problem. They give sufficient conditions for existence, global KKT optimality, uniqueness, explicit allocation formulas, and robustness to objective-estimation error. They do not prove that an arbitrary target is reachable, decidable, or eventually settled. The central empirical task is calibration: determine whether tensor-derived measurements predict actual reductions in verified settlement cost on held-out proof oceans.

11 Conclusion

A consistent settlement optimizer must keep two layers separate. The terminal layer is discrete, one-hot, and verifier-gated. The navigation layer may be fractional, probabilistic, density-based, tensor-valued, and optimized continuously or discretely. This separation permits aggressive search guidance without weakening the meaning of mathematical settlement.

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